Transcription Factors and Signal Transduction

Focus: Transcription factors as the key controllers of gene expression.
Chapter 4 of Transcription Factors and Signal Transduction, Carsten Carlberg (2024)

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Content

- Introduction to Transcription Factors
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- ChIP-seq
- Classification of Transcription Factors
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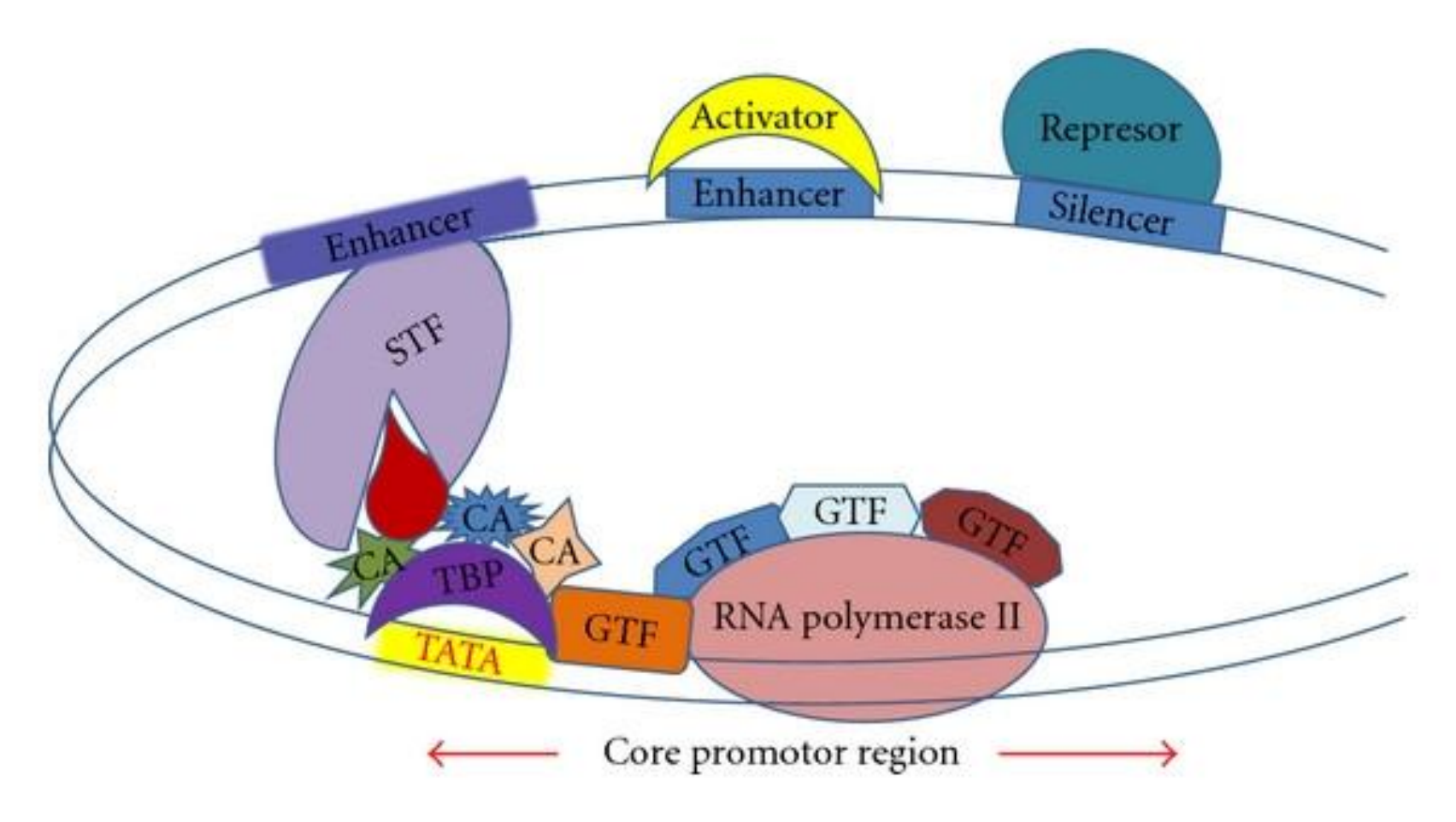
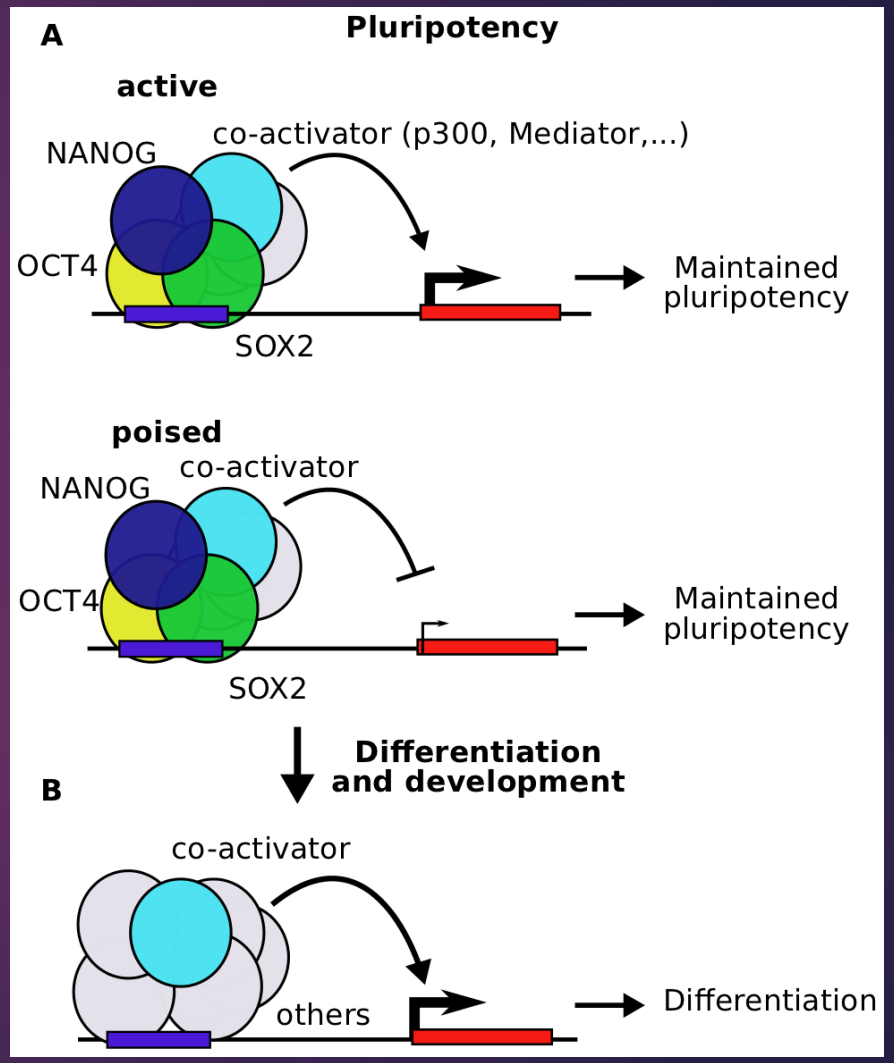
Introduction to Transcription Factors



DOI:10.1155/2011/461979

What are Transcription Factors (TFs)?

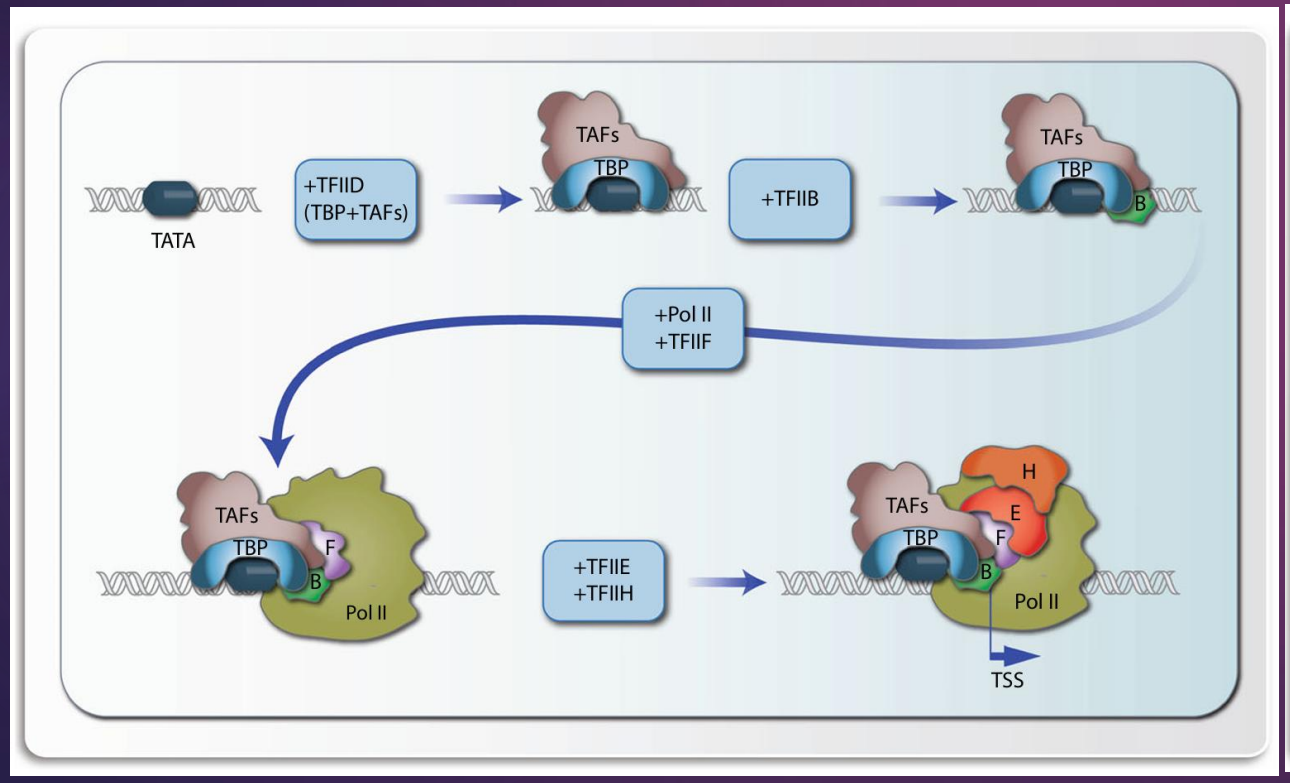
- Definition: Proteins that bind sequence-specifically to genomic DNA to control gene expression.
- Role: Key controllers of cell function and response to environmental perturbations.
- Scale: Humans have ~1,600 TF genes (~8% of all protein-coding genes).
- Mechanism:
  - Bind to Enhancers (proximal or distal).
  - Recruit Coactivators and Chromatin Modifiers.
  - Interact with the Basal Transcriptional Machinery (Pol II complex) via the Mediator complex.

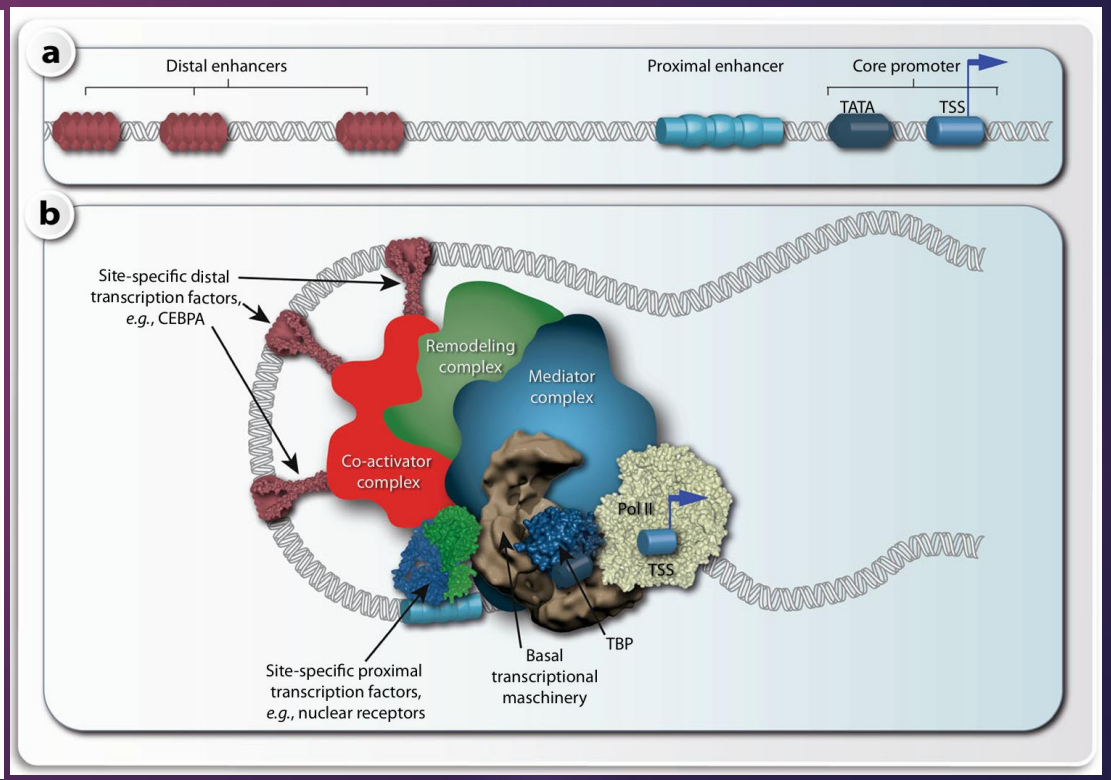


•https://doi.org/10.1371/journal.pcbi.1002304

Site-Specific TFs vs General Transcription Factors (GTFs)

- General Transcription Factors (GTFs): Part of the basal machinery (e.g., TFIID) at the Transcription Start Site (TSS).
- Site-Specific TFs: Regulate specific genes by binding to enhancers.



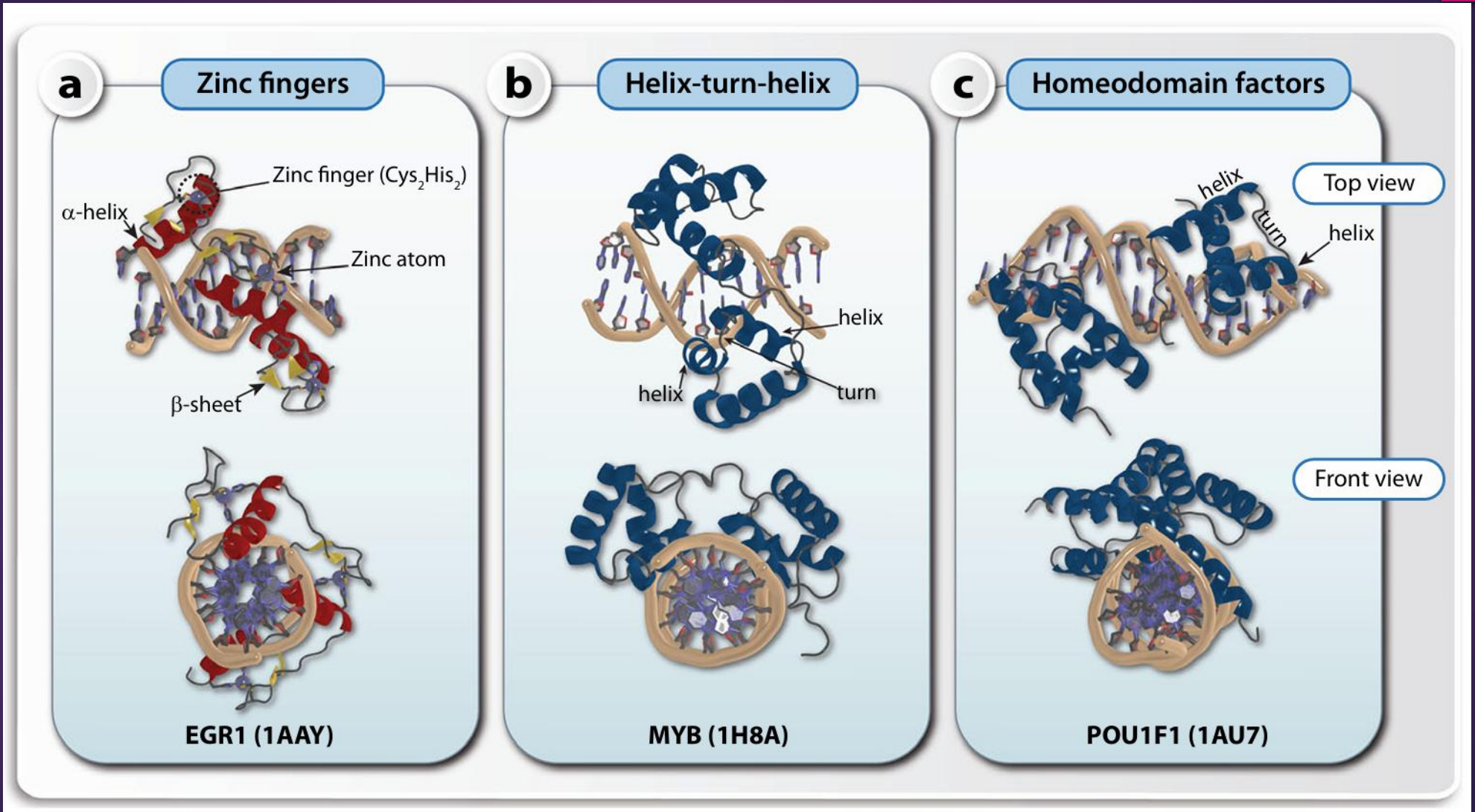


Functional Domains of Transcription Factors

The Modular Structure of TFs A typical TF contains three specific domains:

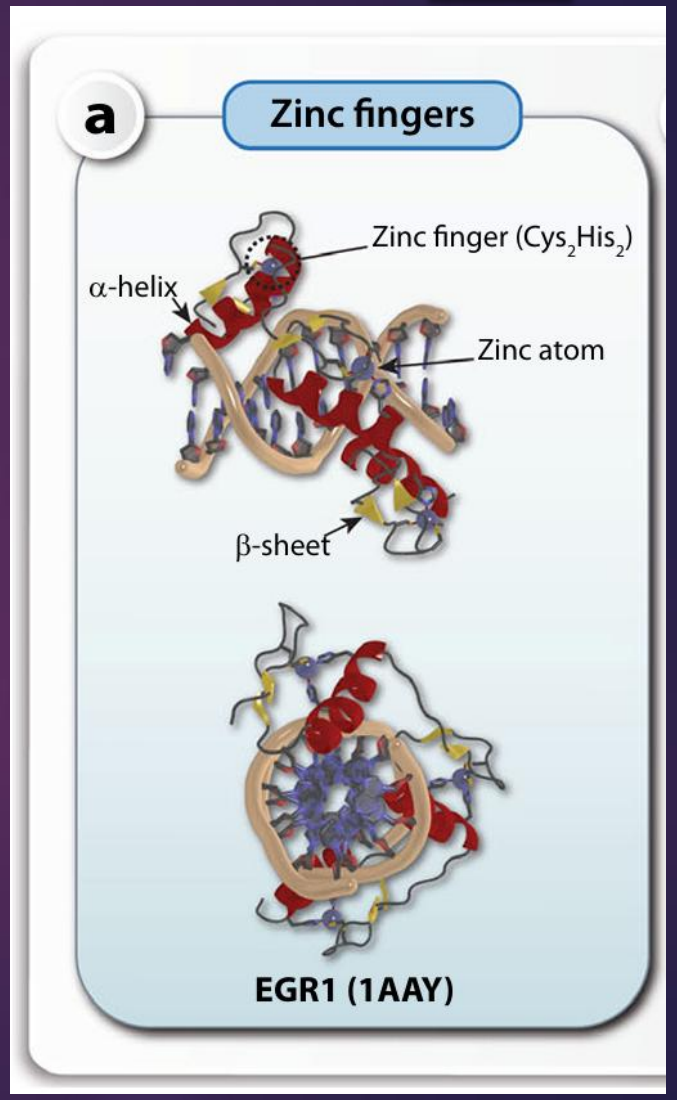
1. DNA-Binding Domain (DBD):
   - Recognizes specific DNA sequences (motifs).
   - Average size: 60–90 amino acids.
   - Interacts primarily with the Major Groove of DNA via hydrogen bonds.
   - Basic amino acids (Lys, Arg) interact with the negatively charged DNA backbone.
2. Dimerization Domain:
   - Allows TFs to form homodimers or heterodimers.
   - Increases diversity and specificity of gene regulation.
3. Transactivation Domain:
   - Site for protein-protein interactions (e.g., with Coactivators or Mediator).
   - Connects the TF to the transcriptional machinery.

DNA-Binding Domains (DBD)



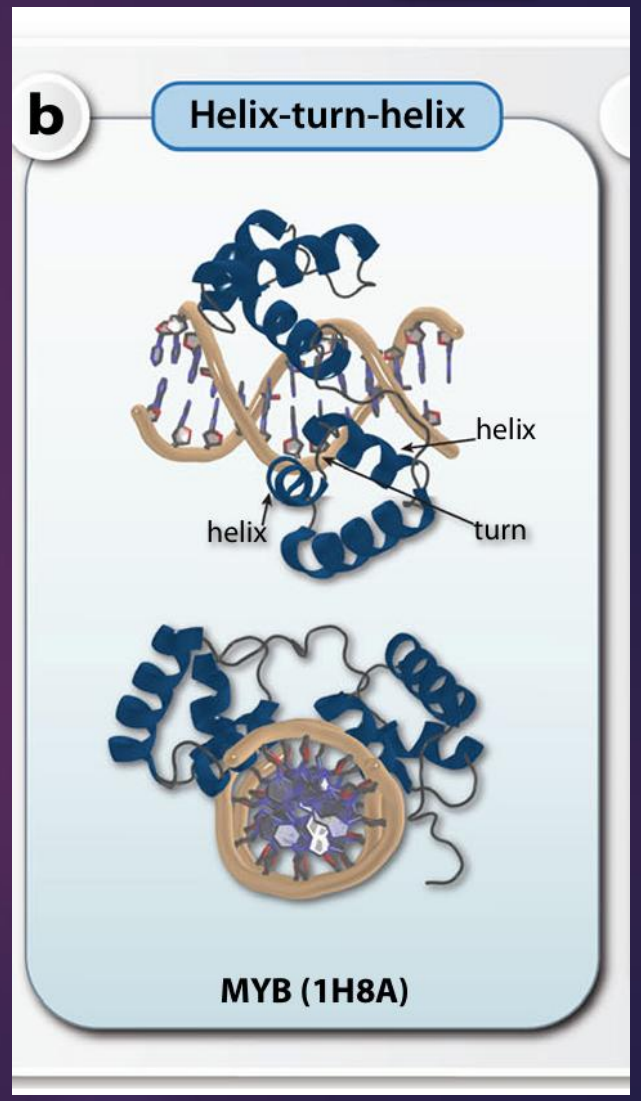
DNA-Binding Domains (DBD) – Class I: Zinc Fingers

- Structure:
  - Small motif (~30 amino acids).
  - Coordinates a Zinc ion (Zn^{2+})using 4 residues:
    - C4 Type: 4 Cysteines (common in Nuclear Receptors).
    - C2H2 Type: 2 Cysteines + 2 Histidines.
- Function:
  - The Zinc ion stabilizes the 3D structure (creates a "finger" loop).
  - Cooperativity: A single finger binds weakly; TFs usually have multiple fingers linked in a chain to bind DNA with high affinity.
  - Example: CTCF (11 zinc fingers), Sp1 (3 zinc fingers).



DNA-Binding Domains (DBD) – Class II: Helix-Turn-Helix

- Helix-Turn-Helix (HTH):
  - Structure: Two alpha-helices separated by a beta-turn.
  - Mechanism: One helix is the "Recognition Helix" that sits in the major groove.
  - Evolution: Ancient motif found in bacteria and eukaryotes.
- Homeodomain (Subtype of HTH):
  - Definition: A specific 60-amino acid HTH motif.
  - Role: Critical for body pattern formation during development.
  - Examples: HOX genes (Homeobox), POU family.

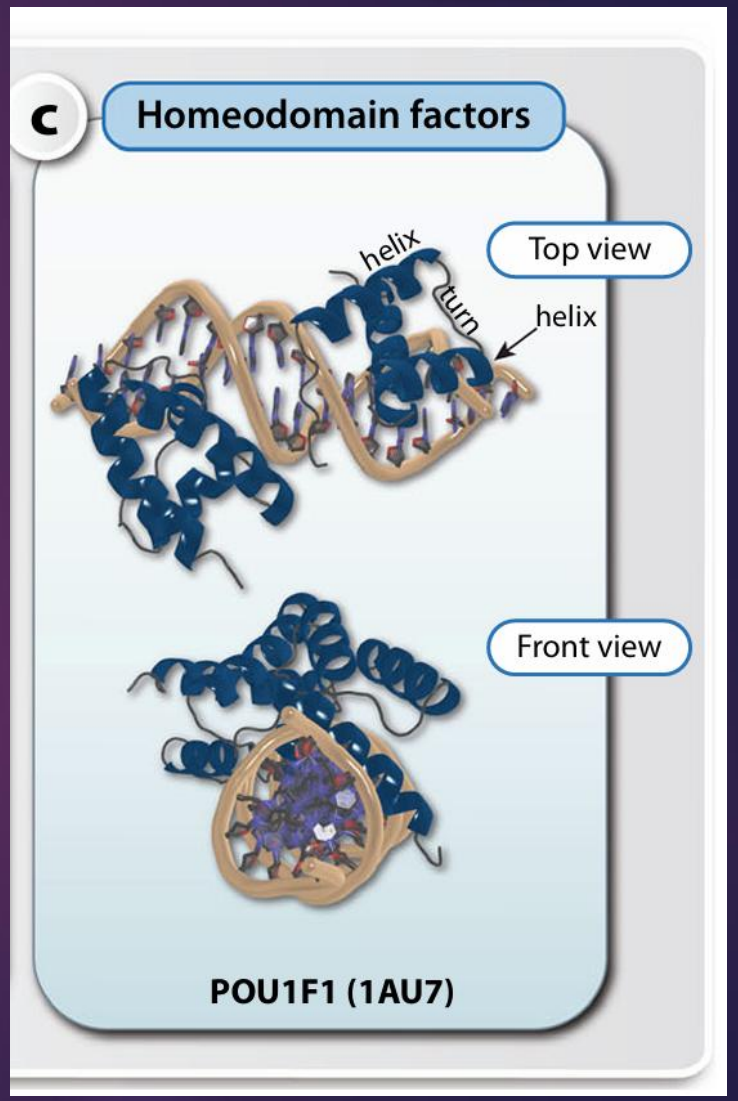


DNA-Binding Domains (DBD) – Class III: Homeodomain factors Dimerization Domains

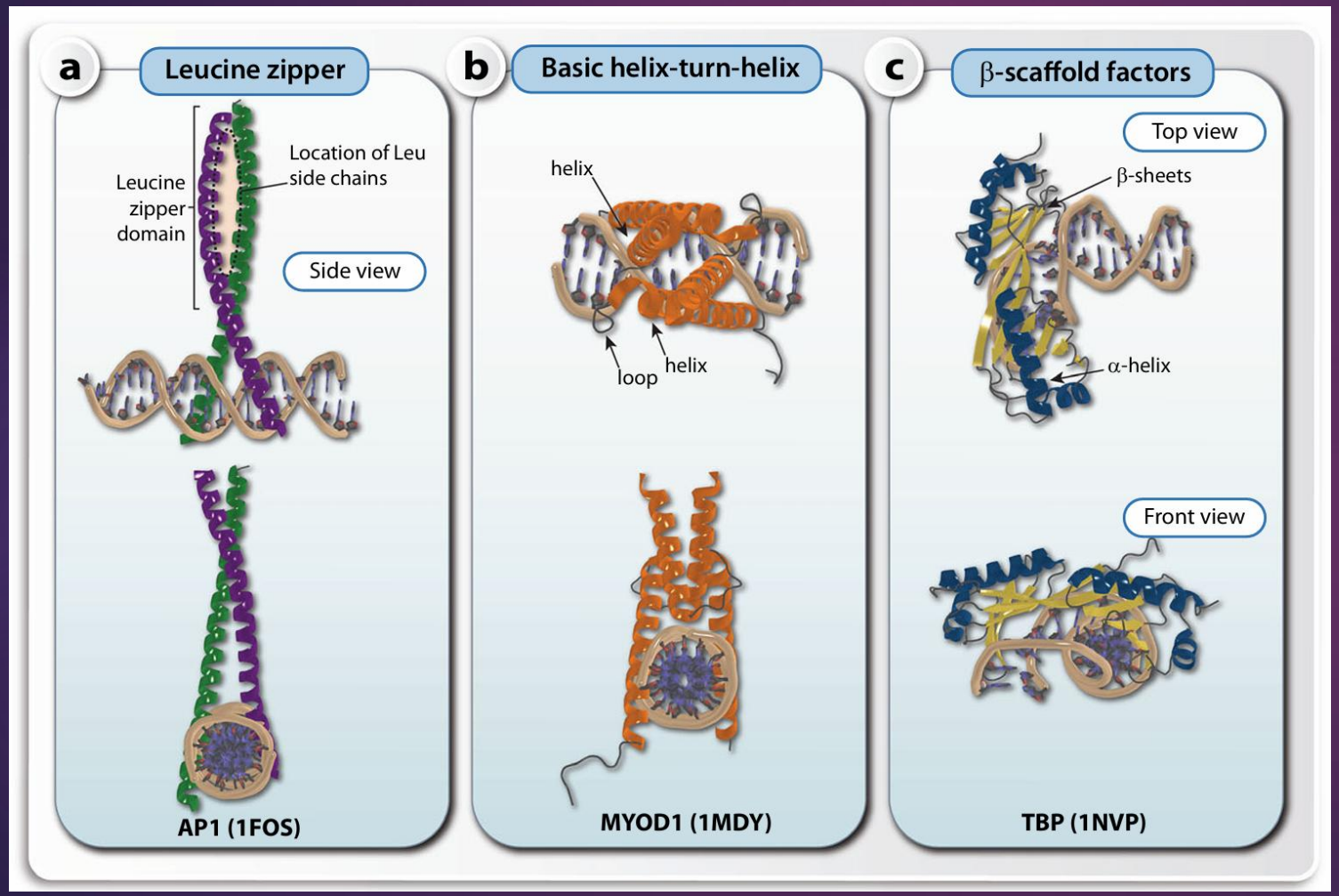
1. Leucine Zipper (bZIP)
- Structure: Two alpha-helices with Leucine residues every 7th position.
- Mechanism: Leucines create a hydrophobic surface that "zips" two helices together.
- DNA Binding: Adjacent basic region binds DNA.
- Example: AP1 (Fos/Jun heterodimer).

2. Basic Helix-Loop-Helix (bHLH)
- Structure: Two helices connected by a flexible loop.
- Mechanism: Loop allows folding; helices mediate dimerization.
- Example: MYOD1 (Muscle differentiation), MYC (Oncogene).

3. Beta-Scaffold (Minor Groove Contacts)
- Mechanism: Distorts/bends DNA by inserting amino acids into the Minor Groove.
- Example: TBP (TATA-binding protein), SOX factors.

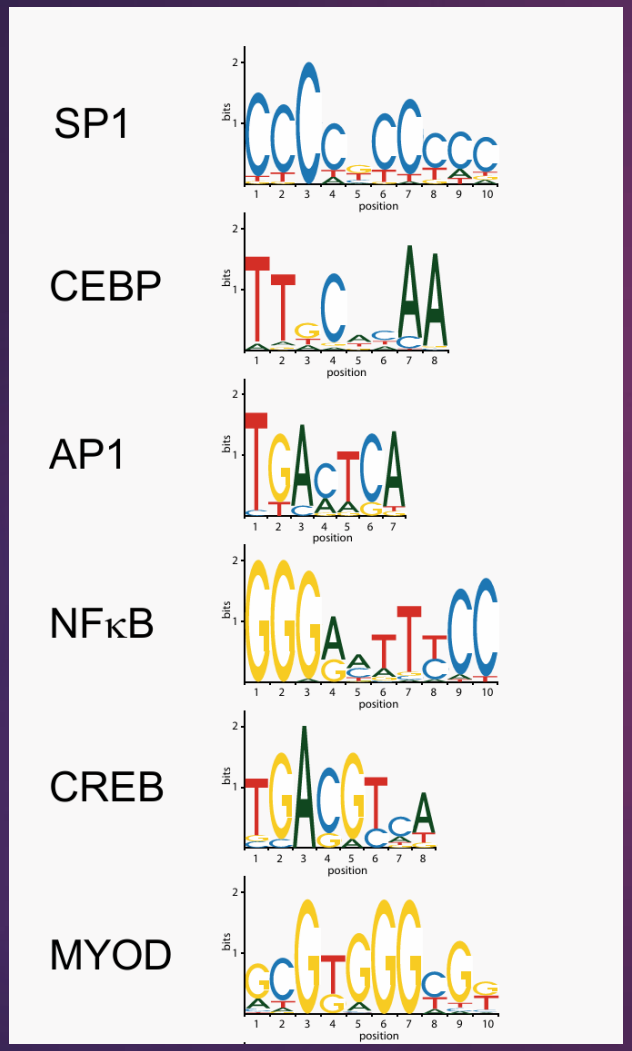


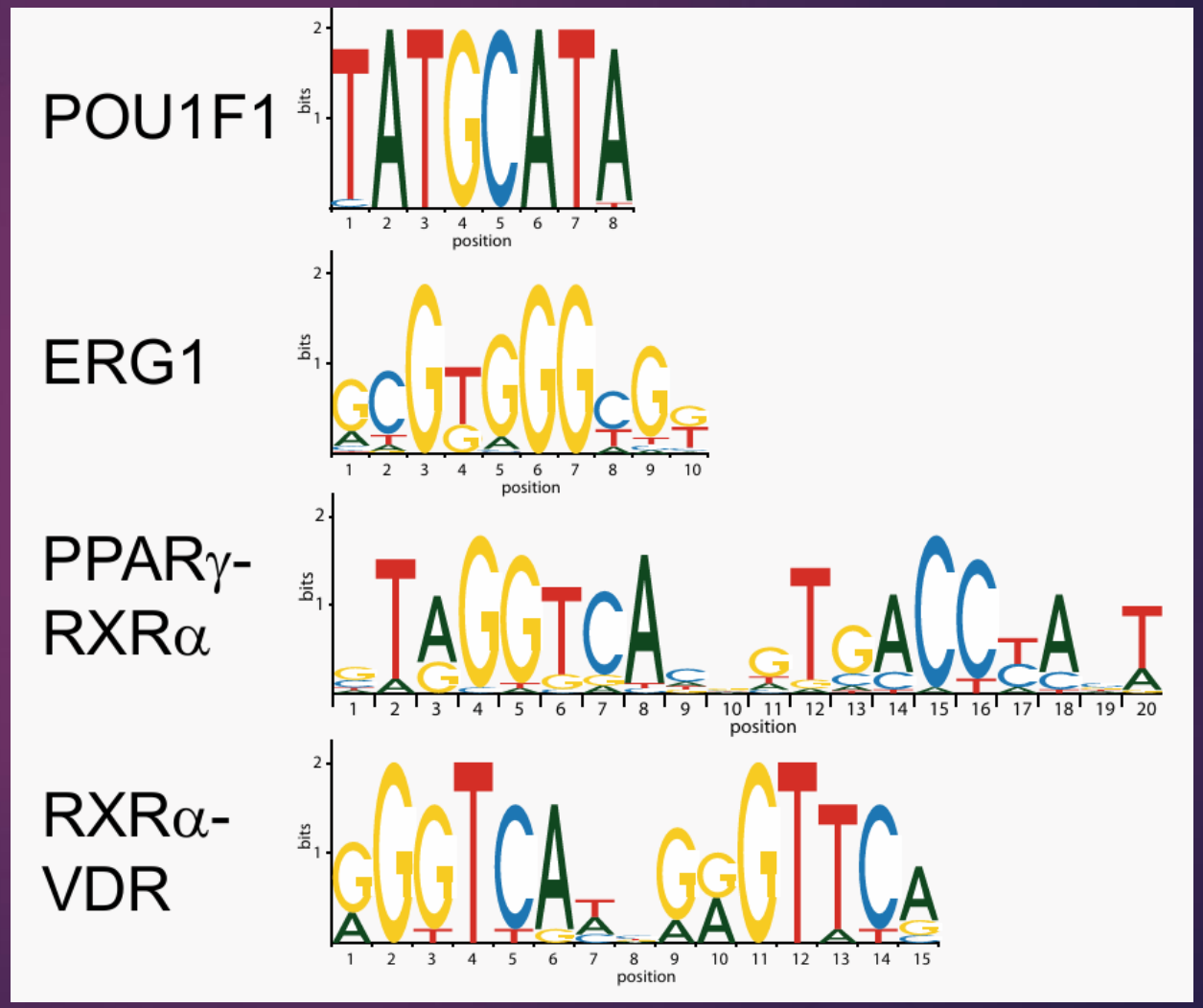
Three main classes of protein–DNA interaction modes of transcription factors



- The DNA interaction of transcription factors is often directed by their mode of dimerization.
- Representatives of the three main groups, leucine zipper a, helix-loop-helix b and β-scaffold factors with minor groove contacts c, are displayed in two different orientations

Sequence logos of key TFBSs





ChIP-seq for Histone modifications, TFBSs and DNAaseI Hypersensitivity

ChIP-seq overview

DNA + bound protein

Fragment DNA

Immunoprecipitate

genomic DNA

POI

POI

cross-link and shear

POI

POI

IP

antibody

POI

POI

Release DNA

Prepare sequencing library

Sequence

Map sequence tags to genome & identify peaks

CHROMOSOMAL DNA

H3K4me3 ChIP-Seq

ChIP-chip

H3K27me3 ChIP-Seq

ChIP-chip

Atp6v0a

Naglu

Hsd17b1

Coasy

Mfix

Psmc3ip

Tubg1

Tubg2

Adapted from slide set by: Stuart M. Brown, Ph.D.,
Center for Health Informatics & Bioinformatics, NYU School of Medicine

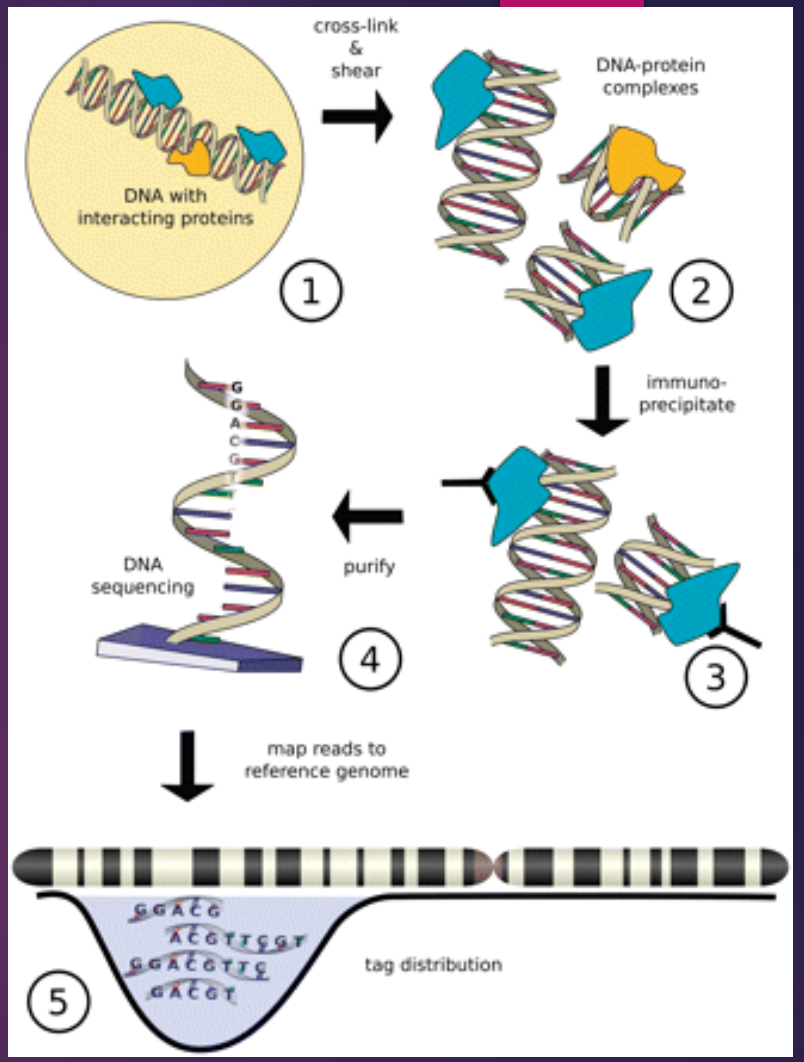
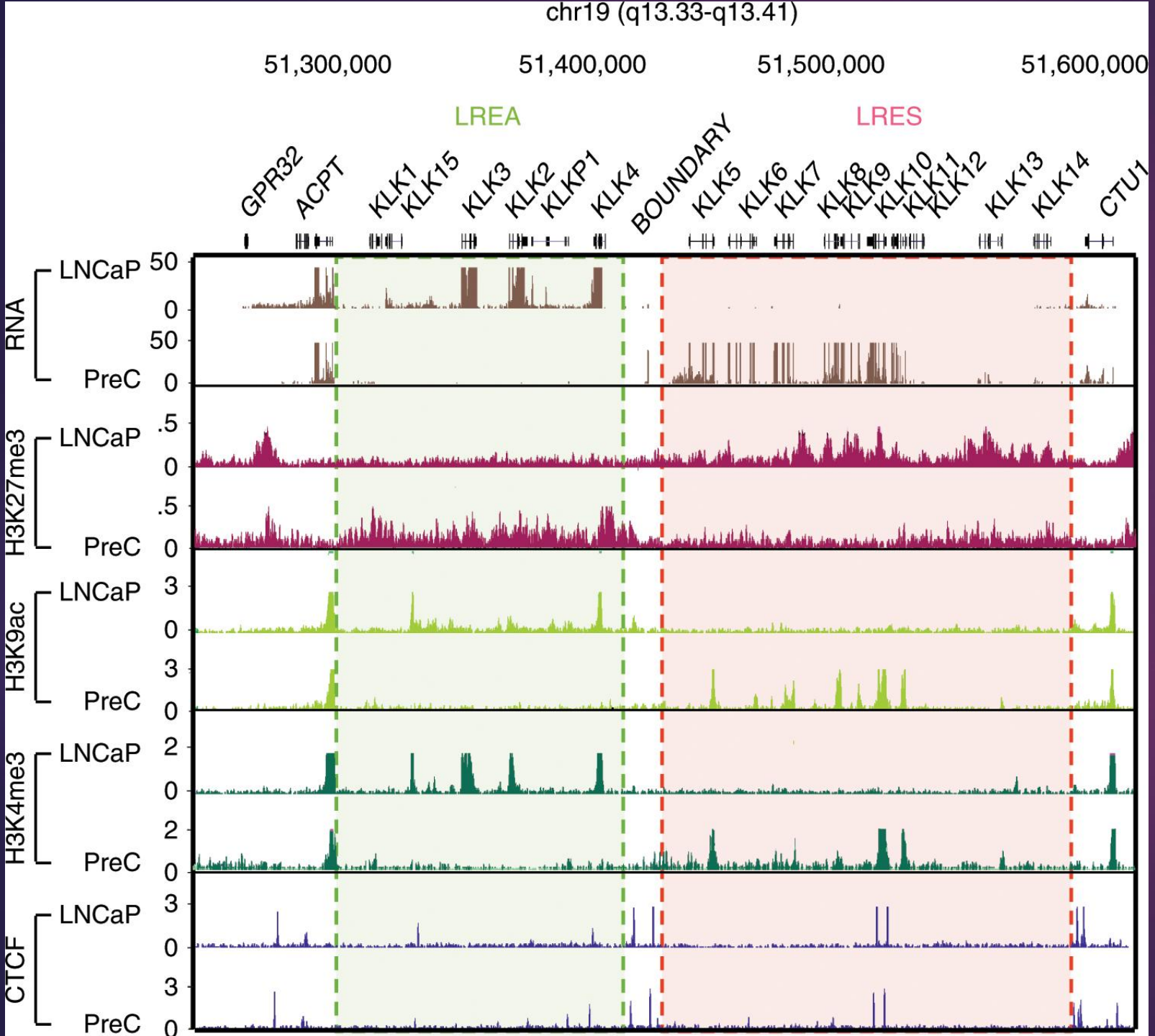
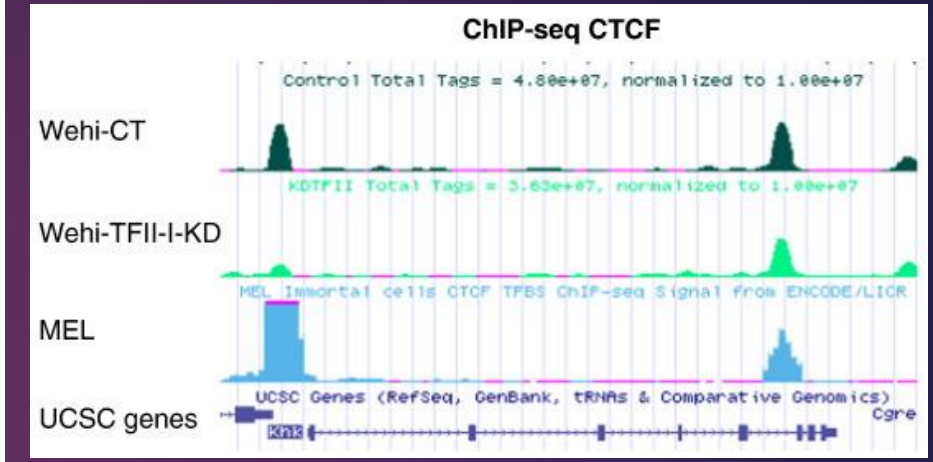


Figure from Szalkowski, A.M, and Schmid, C.D.(2010). *Rapid innovation in ChIP-seq peak-calling algorithms is outdistancing banchmarking efforts.* Briefings in Bioinfomatics.

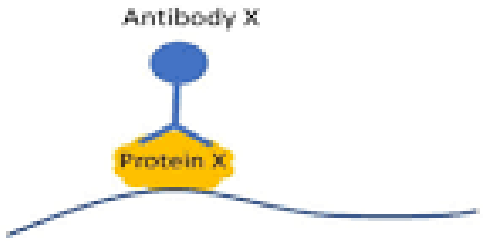




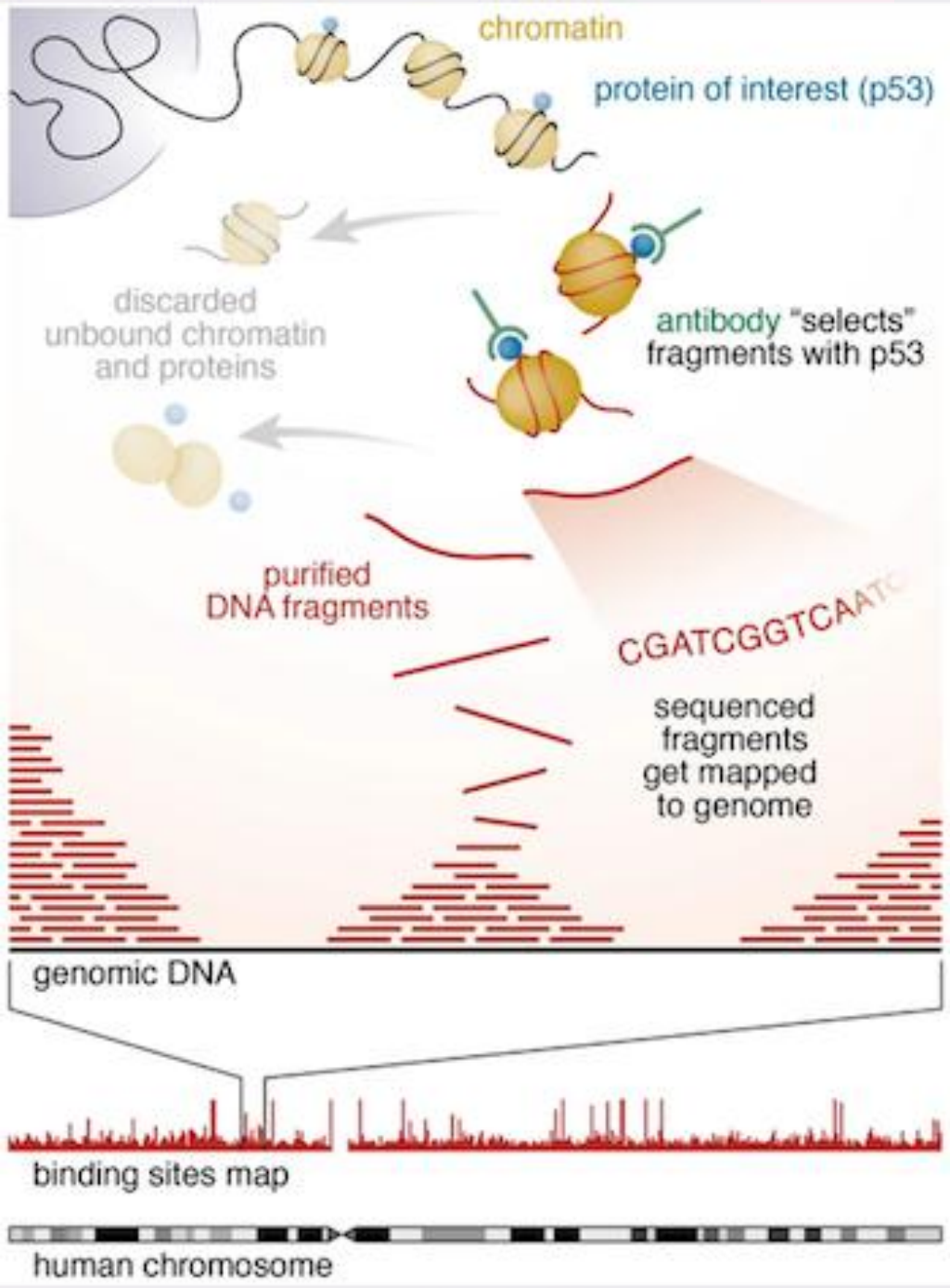
10.1016/j.gdata.2014.09.012

https://www.nature.com/articles/s41467-019-13753-7/figures/1

ChIP seq

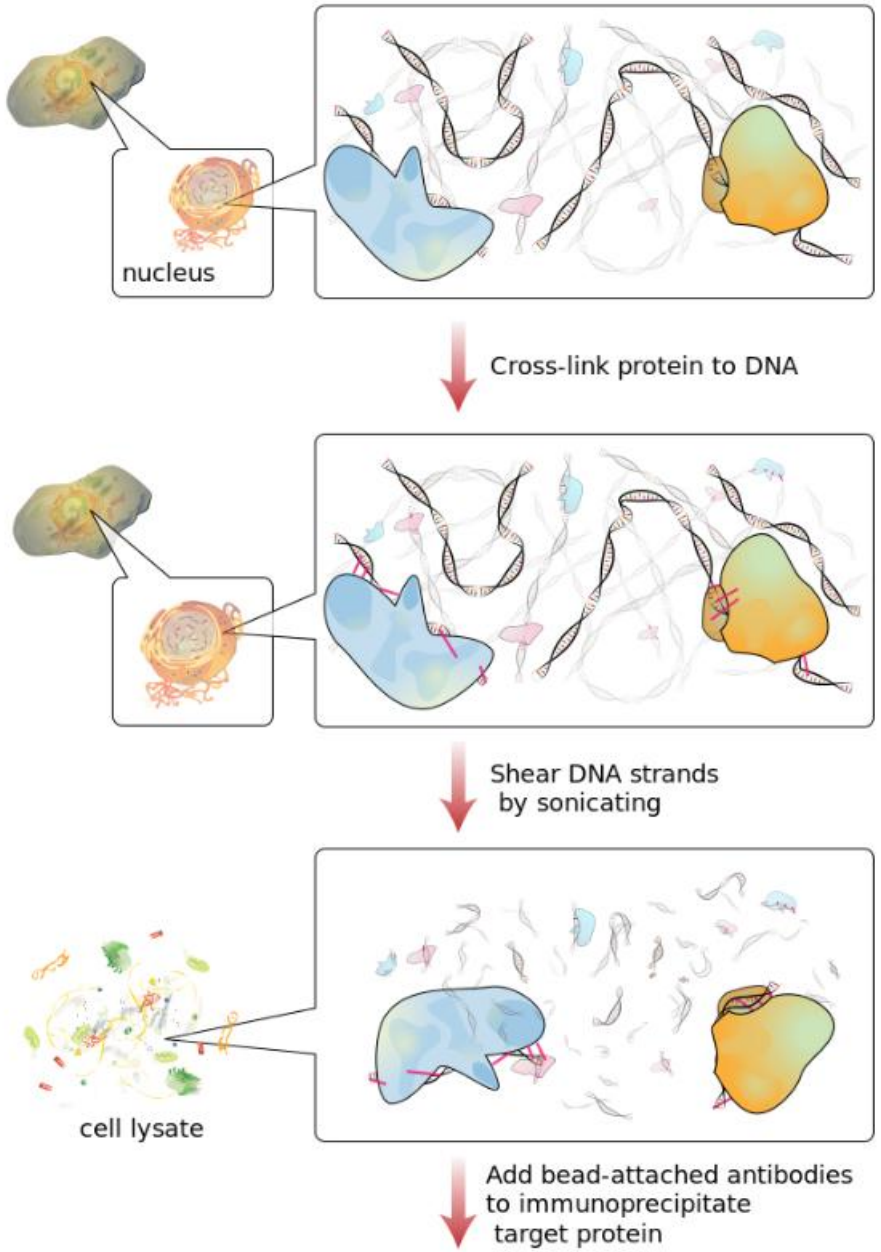


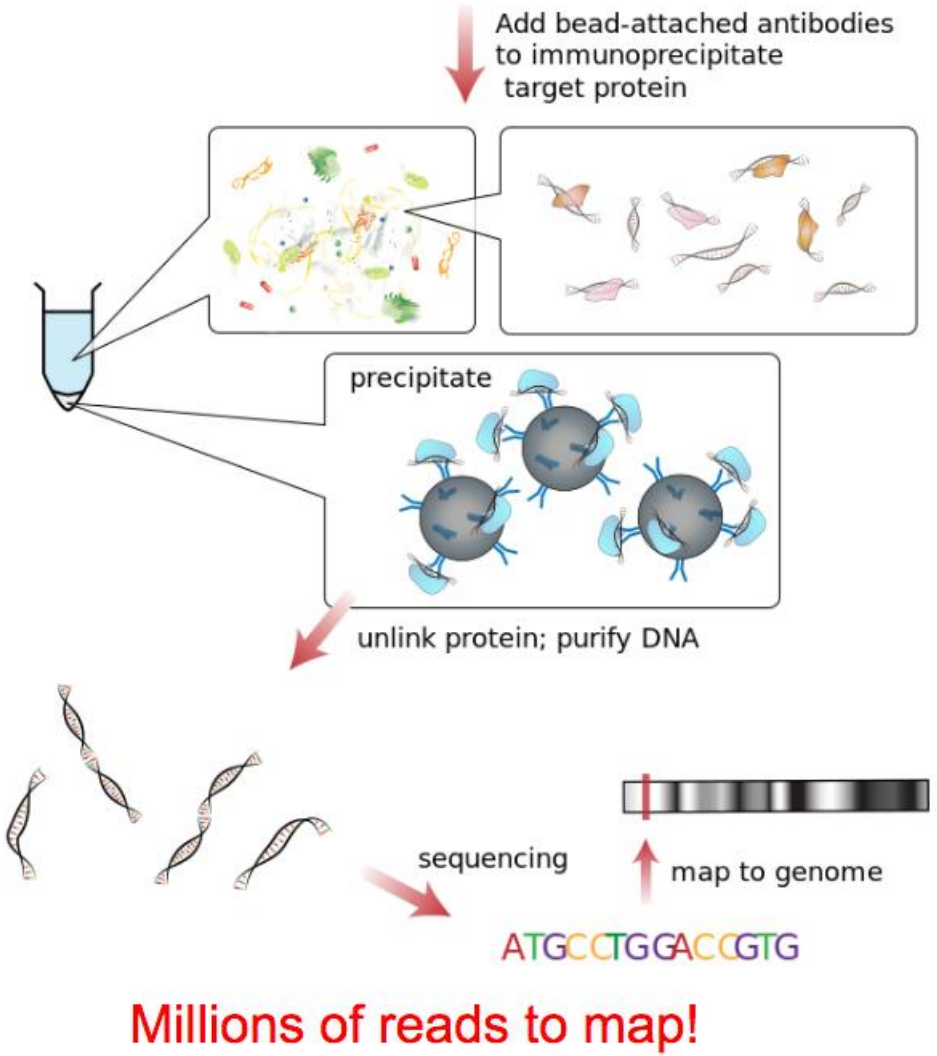
Chromatin immunoprecipitation sequencing



https://hbctraining.github.io/Intro-to-ChIPseq/lessons/01_Intro_chipseq_data_organization.html

Input data is not going through the IP step!

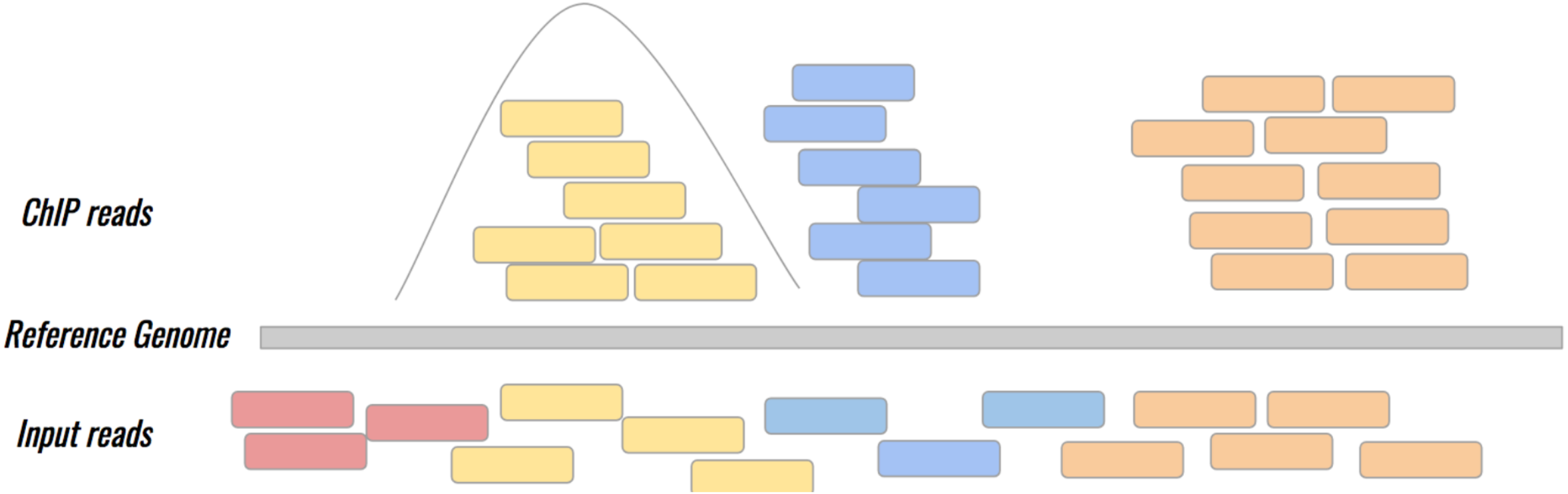




Millions of reads to map!

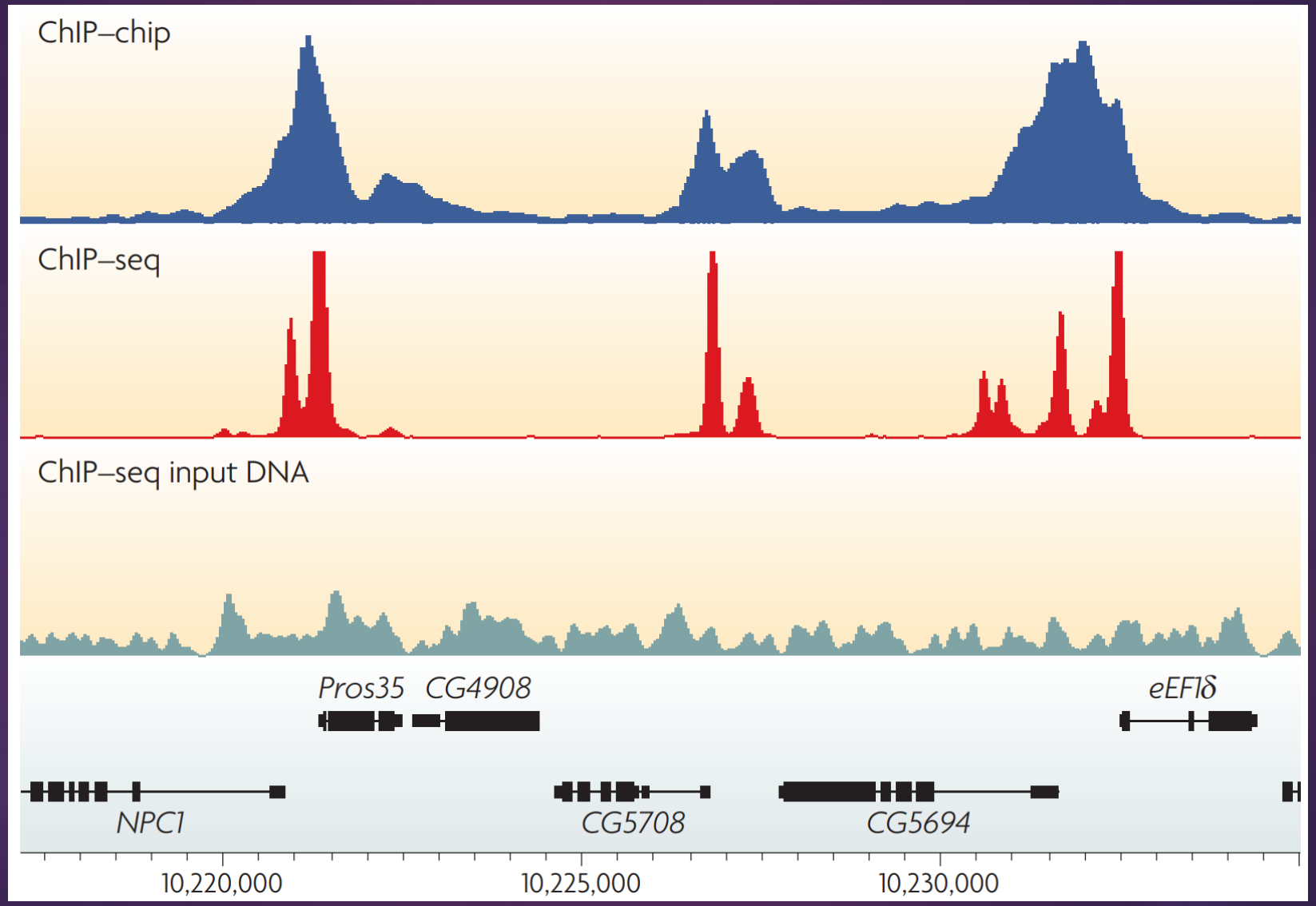
html

ChIP vs Input



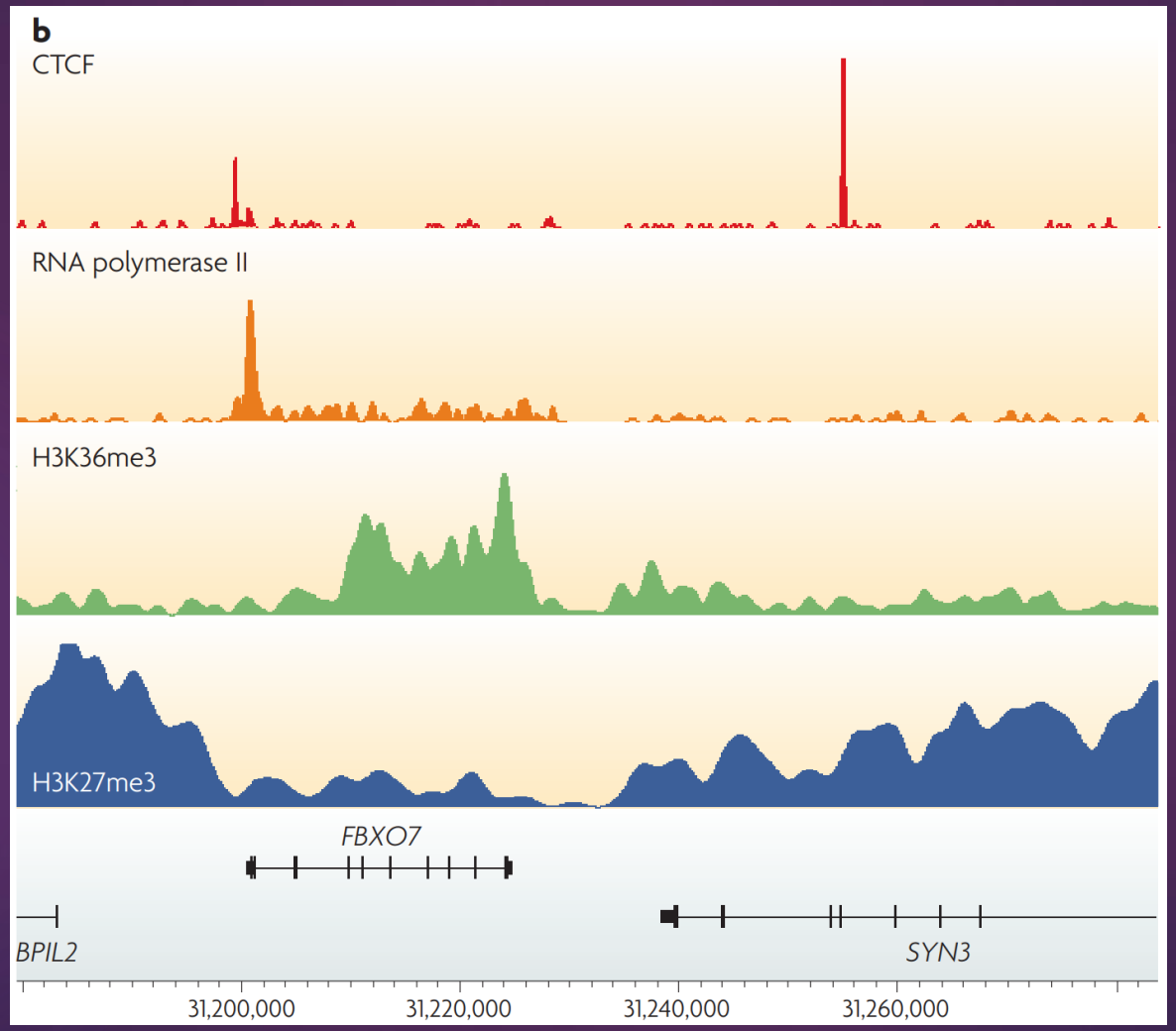
https://training.galaxyproject.org/training-material/topics/epigenetics/tutorials/formation_of_super-structures_on_xi/slides-plain.html

ChIP profiles: Examples of the profiles generated by chromatin immunoprecipitation



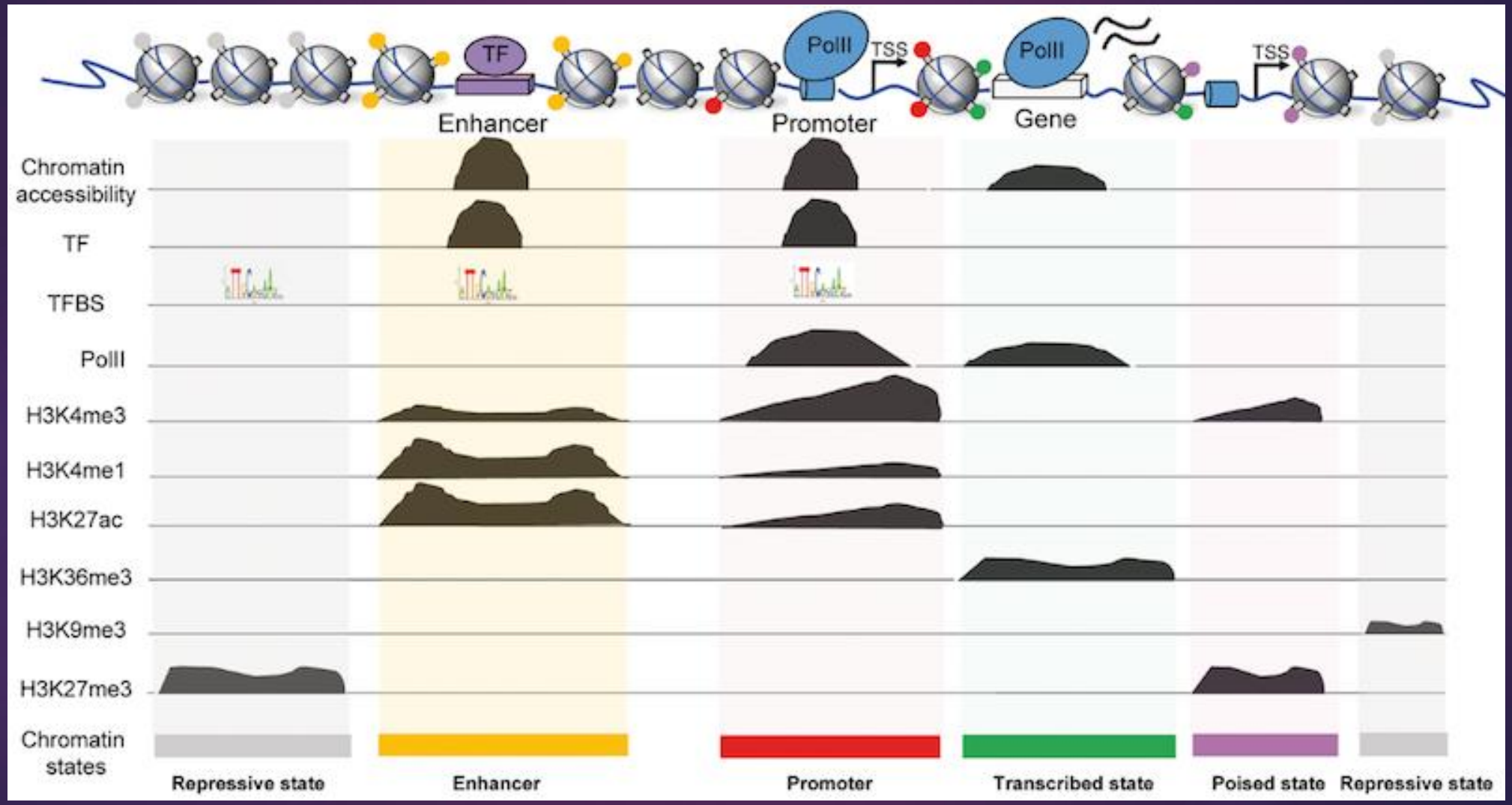
https://www.nature.com/articles/nrg2641

ChIP profiles: Examples of different types of ChIP–seq tag density profiles in human T cells



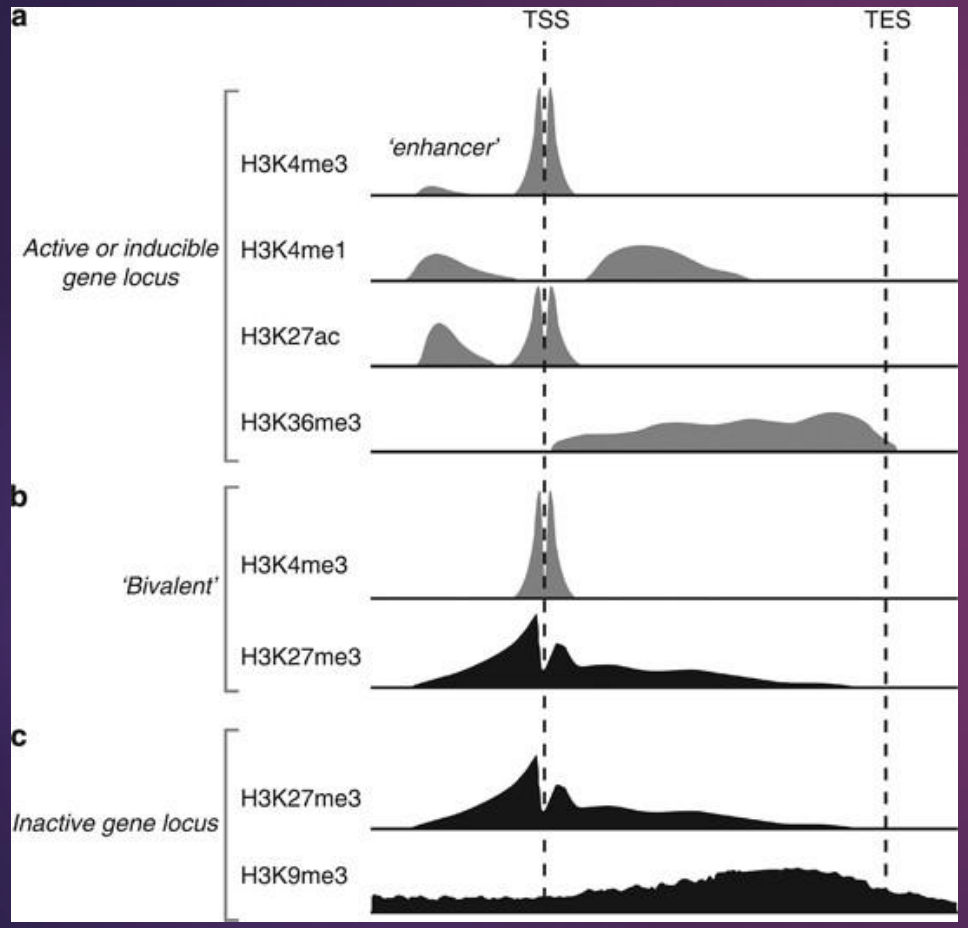
https://www.nature.com/articles/nrg2641

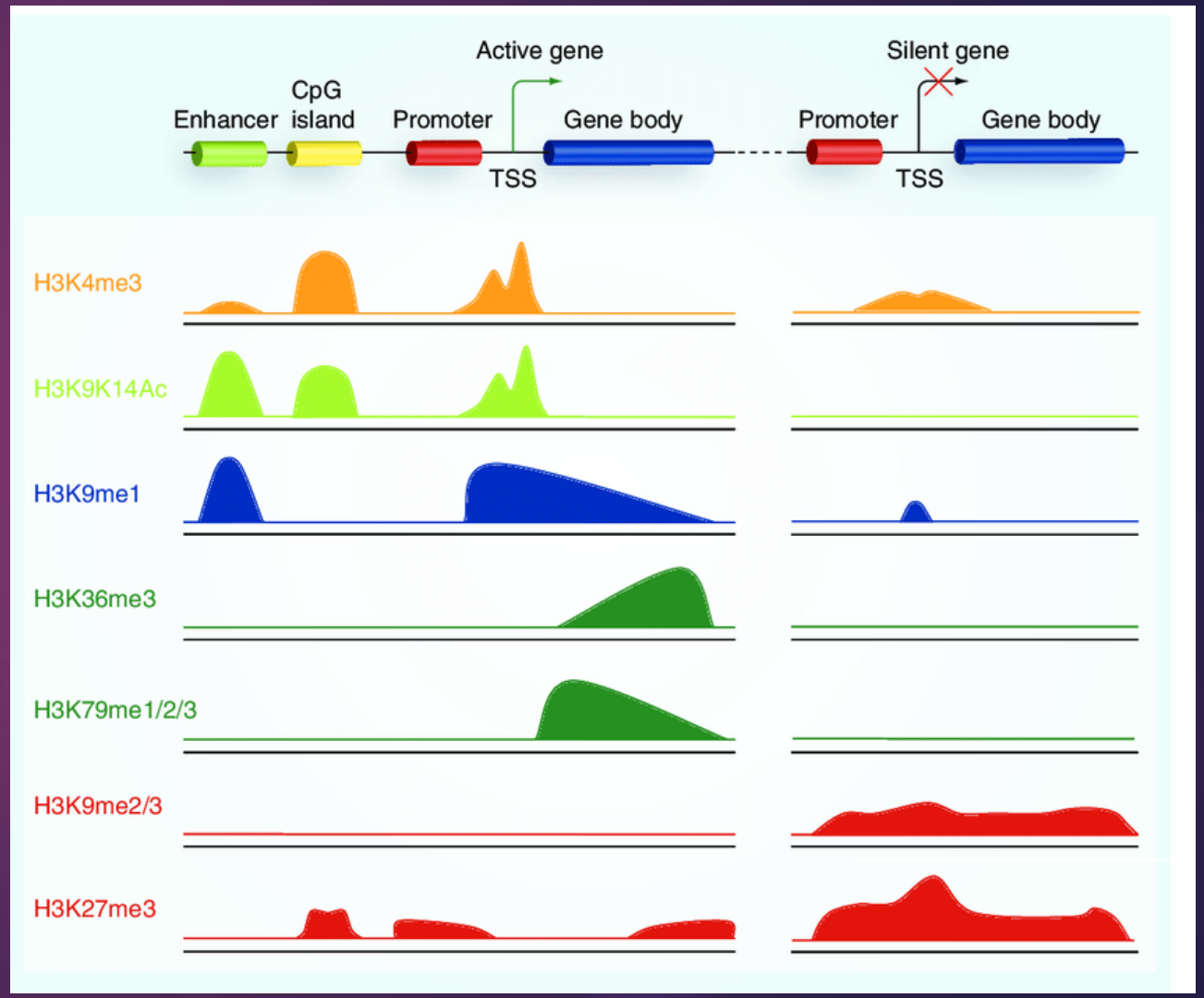
Types of ChIP-seq



https://training.galaxyproject.org/training-material/topics/epigenetics/tutorials/formation_of_super-structures_on_xi/slides-plain.html

Peak types and locations of histone methylation

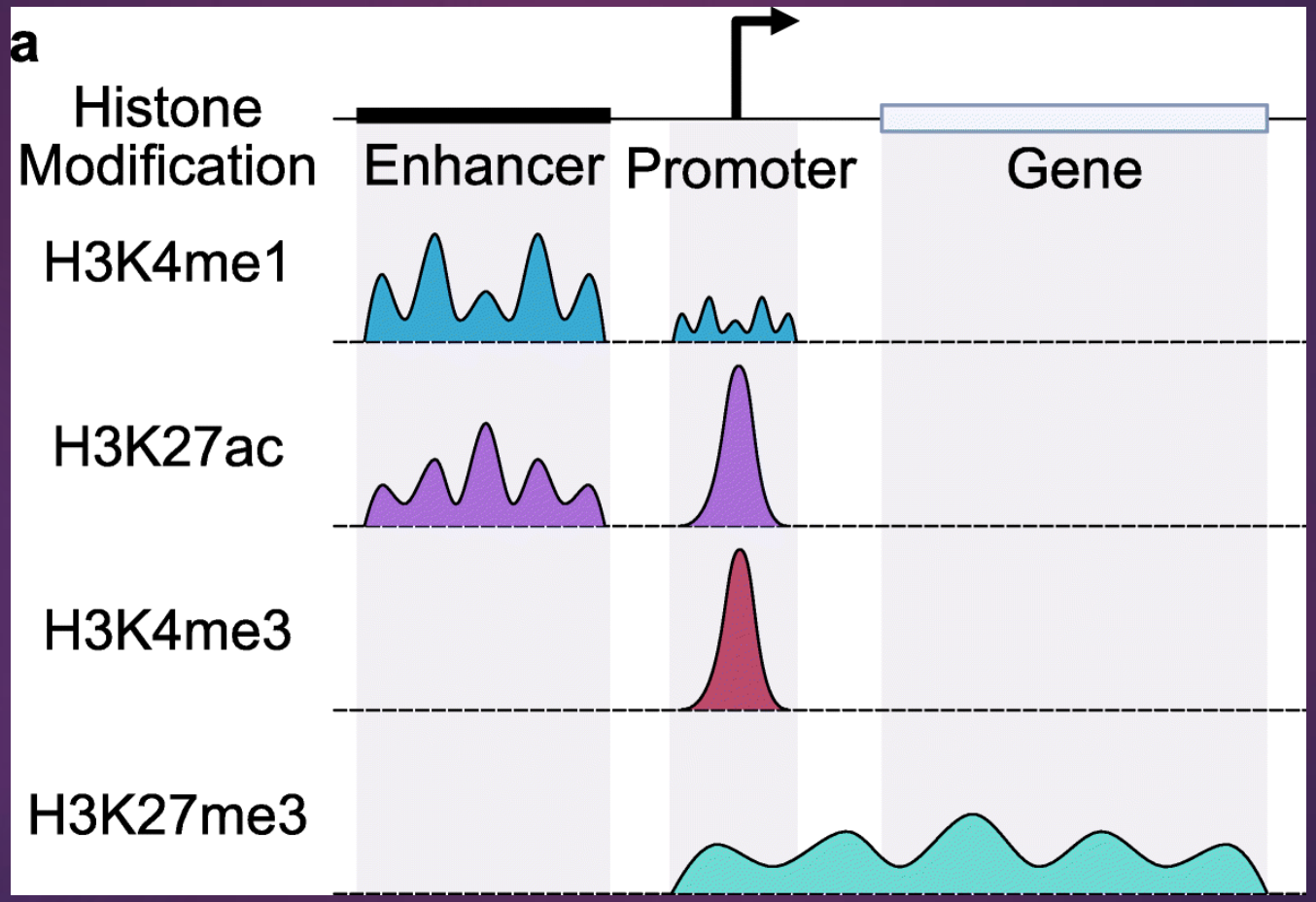




https://www.nature.com/articles/jhg201366

DOI:10.2217/epi.10.55

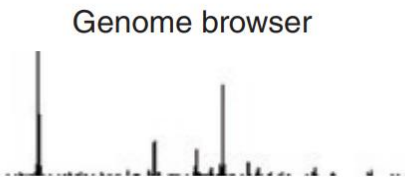
Peak types and locations of histone modifications

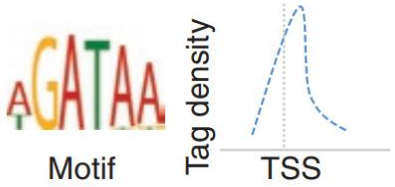


https://genomebiology.biomedcentral.com/articles/10.1186/s13059-022-02707-w

Sequence alignment

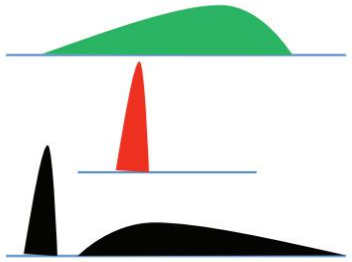
(no longer a bottleneck for data analysis)





Inspect data quality

- Check report from vendor-supplied analytical pipeline
- Visually inspect with a genome browser
- Identify motif in tag-enriched region
- Examine the profile at certain genomic features
- Confirm by quantitative PCR



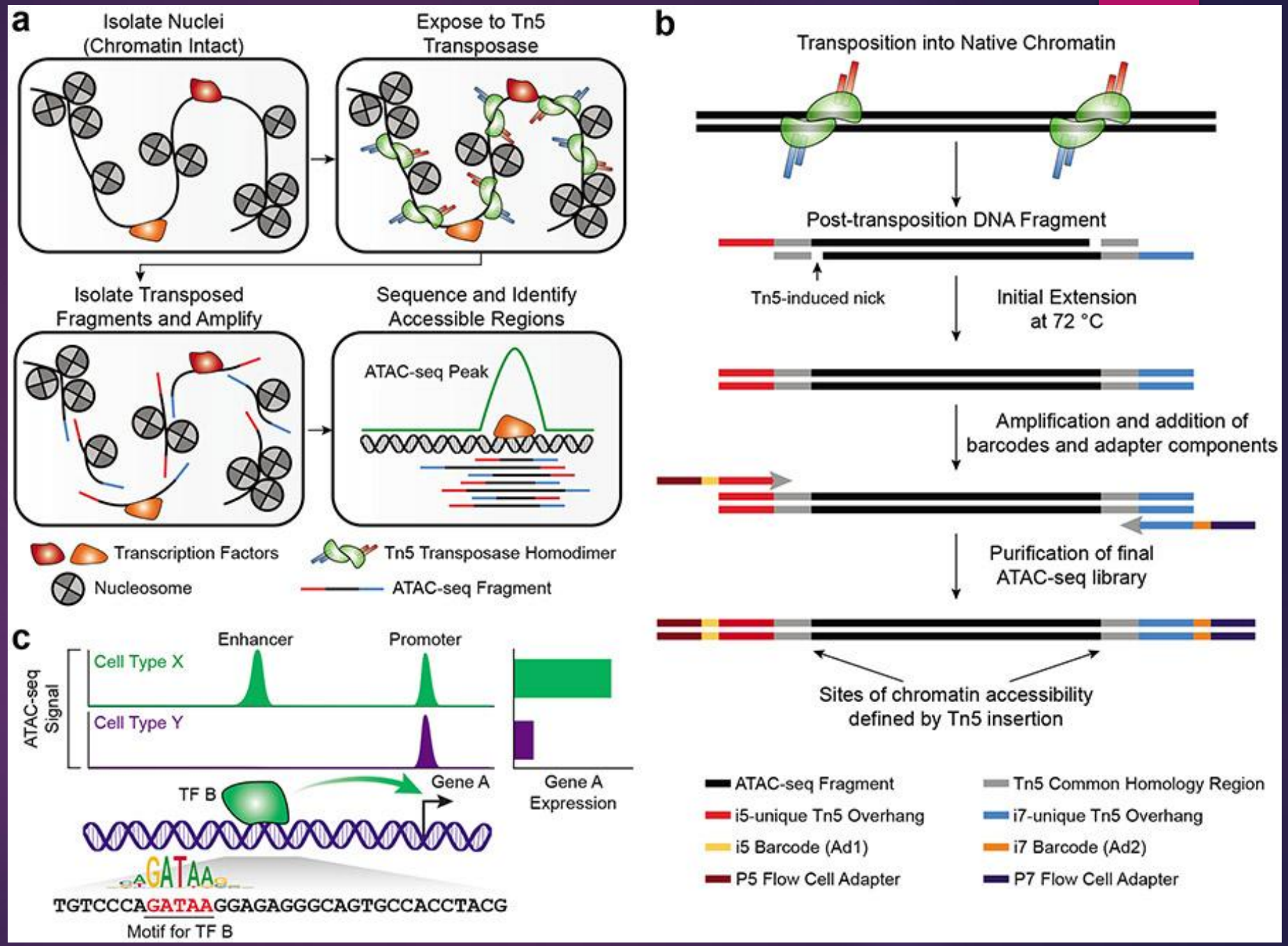
Peak calling (choose the right tool)

Type of peak	Example	Representative tools
Broad	H3K27me3	CCAT, SICER
Sharp	CTCF	MACS
Sharp & broad	Pol II	ZINBA

Figure 2 Common procedures for ChIP-seq data analysis. After base-calling, short-read sequences are aligned to a reference genome. Data quality is assessed by a combination of various strategies, such as visual inspection with a genome browser, motif identification and confirmation by ChIP and quantitative PCR. The initial inspection or prior knowledge provides information about whether the peaks are broad or sharp or both. Various algorithms (bottom right) have been developed for the identification of peaks of these three groups.

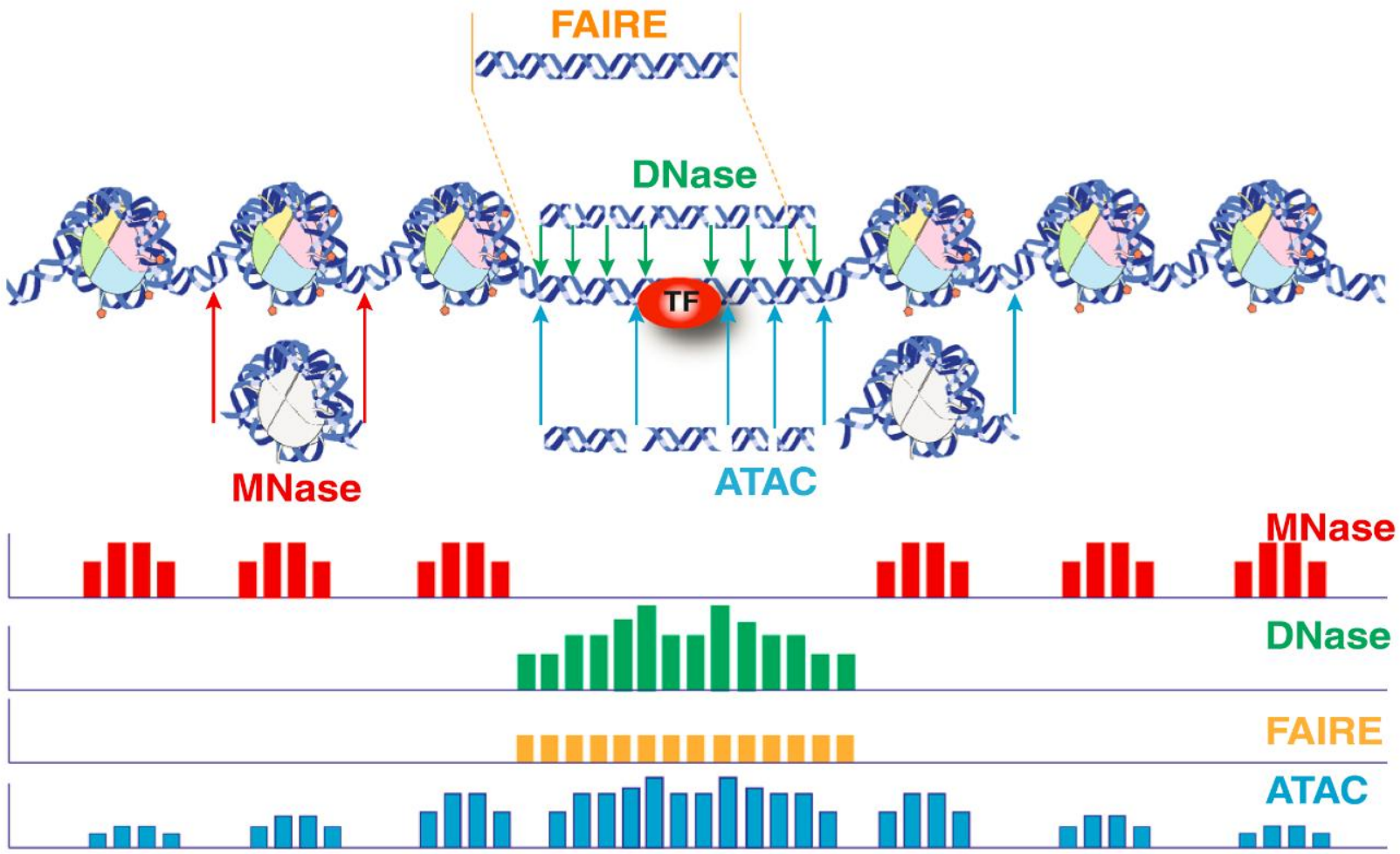
https://www.nature.com/articles/ni.2117

ATAC-seq



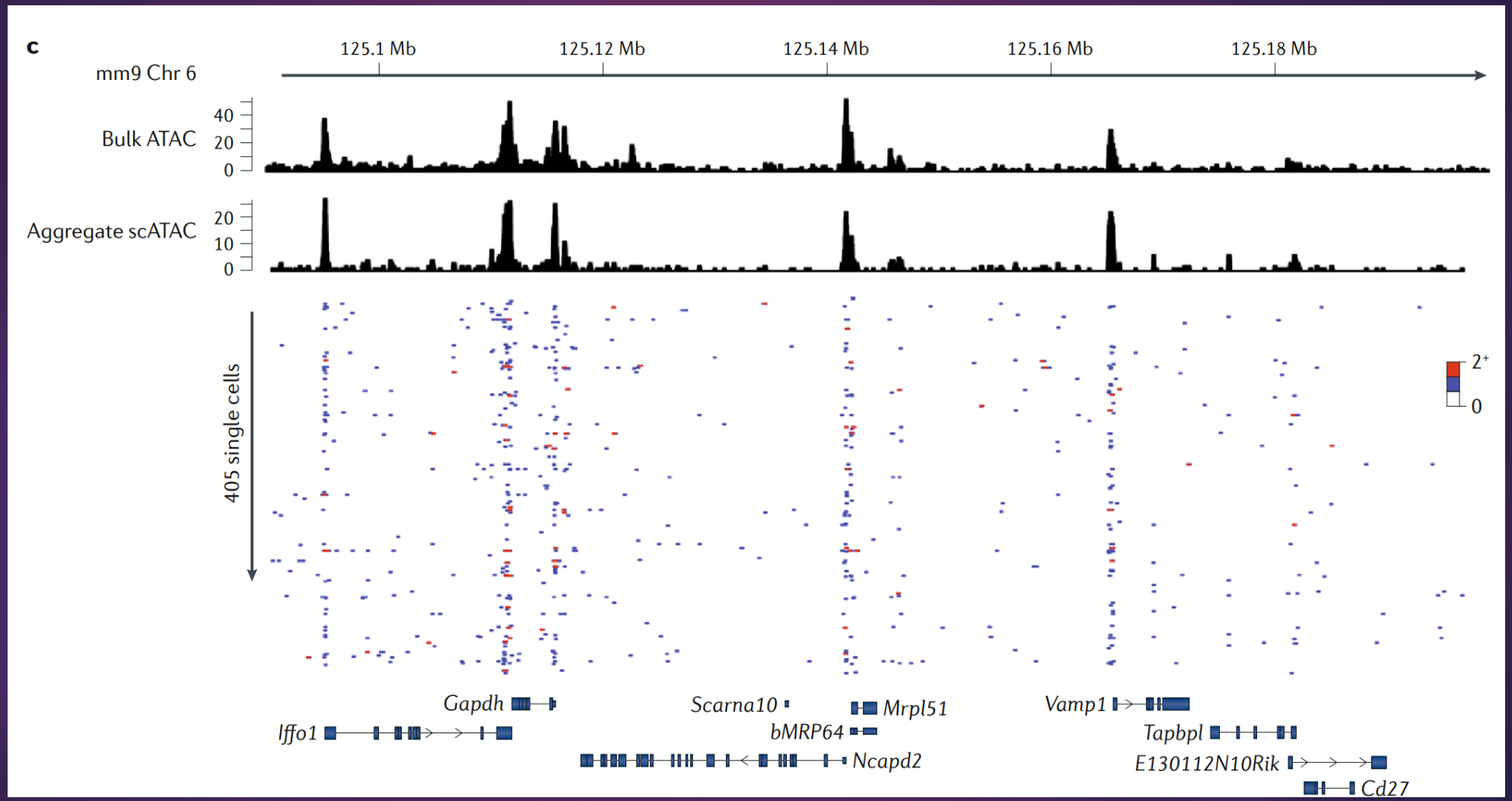
https://pubmed.ncbi.nlm.nih.gov/35478247/

ATAC-seq



https://haibol2016.github.io/ATACseqQCWorkshop/articles/ATACseqQC_workshop.html

Population-scale measurements of chromatin accessibility reflect the average accessibility of a heterogeneous collection of single molecules



https://www.nature.com/articles/s41576-018-0089-8

Classification of Transcription Factors

TFs are classified by how they are activated:

1. Constitutive Factors (Always present/active)
   - Ubiquitous: Present in all cells (e.g., SP1).
   - Cell-Type Specific: Developmental factors (e.g., MYOD).
2. Signal-Dependent Factors (Require activation)
   - Nuclear Receptors: Activated by lipids/hormones.
   - Internal Signals: Activated by metabolites (e.g., p53, SREBP).
   - Cell Surface Receptor Dependent:
     - Phosphorylation: MAPK pathway (e.g., CREB).
     - Latent Cytoplasmic: Translocate upon signal (e.g., NFkB).

Classification of Transcription Factors

Table 4.1 Classification of human transcription factors

#	Superclasses	Classes
I	Basic domains	Basic leucine zipper factors (bZIP) Basic helix-loop-helix factors (bHLH) Basic helix-span-helix factors (bHSH)
II	Zinc-coordinating DBDs	Nuclear receptors with C4 zinc fingers Other C4 zinc finger-type factors C2H2 zinc finger factors C6 zinc cluster factors DM-type intertwined zinc finger factors CXXC zinc finger factors C2HC zinc finger factors C3H zinc finger factors C2CH THAP-type zinc finger factors
III	Helix-turn-helix domains	Homeodomain factors Paired box factors Fork head/winged helix factors Heat shock factors Tryptophan cluster factors TEA domain factors ARID domain factors
IV	Other all-α-helix DBDs	High-mobility group (HMG) domain factors Heterodimeric CCAAT-binding factors
V	α-helix exposed by β-structures	MADS box factors E2-related factors SAND domain factors
VI	Immunoglobulin fold	Rel homology region (RHR) factors STAT domain factors p53 domain factors Runt domain factors T-box factors NDT80 domain factors Grainyhead domain factors
VII	β-Hairpin exposed to an a/b-scaffold	SMAD/NF-1 DNA-binding domain factors GCM domain factors
VIII	β-Sheet binding to DNA	TATA-binding proteins A.T hook factors
IX	β-Barrel DBDs	Cold-shock domain factors
X	Not yet defined DBDs	AXUD/CSRNP domain factors NonO domain factors Leucine-rich repeat flightless-interacting proteins NFX1-type putative zinc finger factors GTF2I domain factors CG-1 domain factors Uncharacterized

This classification is based on information of the database TRANSFAC (www.edgar-wingender.de/huTF_classification.html)

Signal-Dependent Activation Mechanisms (I)

1. Endocrine Nuclear Receptors:
   - Ligands: Lipophilic hormones (Estrogen, Testosterone, Cortisol, Vitamin D).
   - Mechanism: Ligand diffuses through membrane → Binds Receptor → Receptor binds DNA.
   - Example: Estrogen Receptor (ESR1).
2. Kinase Signaling Cascades (Phosphorylation):
   - Trigger: Hydrophilic signals (Peptides, Growth Factors) bind membrane receptors (GPCRs, RTKs).
   - Mechanism: Phosphorylation cascade (e.g., MAPK) targets resident nuclear TFs.
   - Example: CREB1 (cAMP response), ELK1, AP1.

Signal-Dependent Activation Mechanisms (II)

3. Latent Cytoplasmic Factors:
   - State: Inactive in the cytoplasm (often held by an inhibitor or membrane).
   - Activation: Signal causes release/cleavage → Translocation to Nucleus.
   - Key Pathways:
     - NF–kappa B: Inflammation.
     - STAT: Cytokine signaling.
     - SMAD: TGF-beta signaling.
     - NOTCH: Developmental signaling.

Specific Pathways: SMAD and STAT

- SMAD Pathway (TGF-beta Family):
  - Signal: TGF-beta, BMPs, Activins.
  - Mechanism: Receptor phosphorylates R-SMADs (SMAD 2/3).
  - Complex: R-SMADs dimerize with SMAD4 (Co-SMAD).
  - Action: Translocate to nucleus → Regulate growth/differentiation.
- STAT Pathway (JAK-STAT):
  - Signal: Cytokines (Interleukins, Interferons).
  - Mechanism: Cytokine binds receptor → JAK kinase phosphorylates STAT.
  - Action: Phosphorylated STATs dimerize → Translocate to nucleus.
  - Relevance: Vital for immune cell regulation.

Specific Pathways: NF-kappa B and WNT

- NF-kappa B Pathway (Inflammation):
  - Resting State: NF-kappa B held in cytoplasm by inhibitor I-kappa B.
  - Signal: TNF, IL1 beta, Bacterial LPS.
  - Activation: IKK kinase phosphorylates I-kappa B → I-kappa B degraded.
  - Result: NF-kappa B (p50/p65) is free to enter nucleus.
- WNT Pathway:
  - Resting State: beta-catenin is degraded by GSK3 complex.
  - Signal: WNT binds Frizzled receptor.
  - Activation: Dishevelled (DVL) blocks GSK3 → beta-catenin accumulates.
  - Result: beta-catenin enters nucleus → Activates TCF7L2 transcription factor.

Specific Pathways: Hedgehog, NOTCH, NFAT

- Hedgehog (Hh):
  - Ligand (Hh) relieves repression of Smoothened receptor.
  - Activates GLI transcription factors.
  - Crucial for embryogenesis.
- NOTCH:
  - Contact-dependent signaling (Delta/Jagged ligands).
  - Mechanism: Proteolytic cleavage of NOTCH receptor.
  - Result: Intracellular Domain (NICD) travels to nucleus → Binds RBPJ.
- NFAT (Ca^{2+} Signaling):
  - Signal: Calcium influx (e.g., in T-cells).
  - Mechanism: Calcineurin (phosphatase) dephosphorylates NFAT.
  - Result: Nuclear translocation.

Clinical Relevance

- Diseases:
  - Cancer: Mutations in p53 (Guardian of the genome) or overexpression of MYC.
  - Developmental Disorders: Mutations in HOX or PAX genes.
  - Immune Disorders: Dysregulation of NF-kappa B or STAT.
- Therapy: TFs are major drug targets (e.g., Tamoxifen targets Estrogen Receptor).

Summary

- TFs convert environmental/internal signals into gene expression changes.
- Classified by structure (Zinc Finger, HTH) and activation mode (Constitutive vs. Inducible).
- Operate in complex networks involving homo/heterodimerization.

Xin chân thành cảm ơn!

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